

Day 4: A Database and Git

Goal: Students replace the in-memory notes with a real database (SQLite), so that the notes stay after the server restarts. Then they save their project with Git and put it on GitHub, without the private `.env` file. Fast finishers can try an optional PostgreSQL track.

Folder	Files
notes/	Day-4-Theory (read before class)
lab/	Day-4-Lab-Sheet (Parts 0 to 9, with the optional Track B as Part 8)
help/	Day-4-Stuck-Sheet
starter/	Everything finished up to Day 3 (<code>notes-board</code> with an in-memory <code>data.js</code> and the Express <code>server.js</code>), plus <code>hello-server</code> and <code>setup-check.js</code>
done/	<code>notes-board</code> with SQLite in <code>data.js</code> , <code>server.js</code> with <code>init()</code> , optional- <code>postgres/data.js</code> , <code>.env.example</code> and <code>.gitignore</code>

Risks of the day

1. **Built-in SQLite (`node:sqlite`) on Windows.** It ran on Node.js 24.21.0 on Linux without a flag and without a warning. It is **not tested** on Windows. Do Pilot C in `../INSTRUCTOR-PILOT-CHECKLIST.md` before this class.
2. **Signing in to GitHub from Git.** This is **not tested** on the lab PCs. Do Pilot D1 before this class. Students need a working GitHub account.
3. **Committing `.env` by mistake.** The lab has a check for this (Part 7). Do not skip it.

Track B (optional)

PostgreSQL is only for fast finishers. Day 5 works without it.

Homework for Day 5

1. Make sure your GitHub account works. Sign in at github.com and write down your username. **You need your GitHub login on Day 5.**
2. Read the hosting reading list at the end of the Day 4 Theory.
3. Run your Notes Board once at home or in the lab, and check that your notes stay after a restart.
4. Write down at least two questions for Day 5.